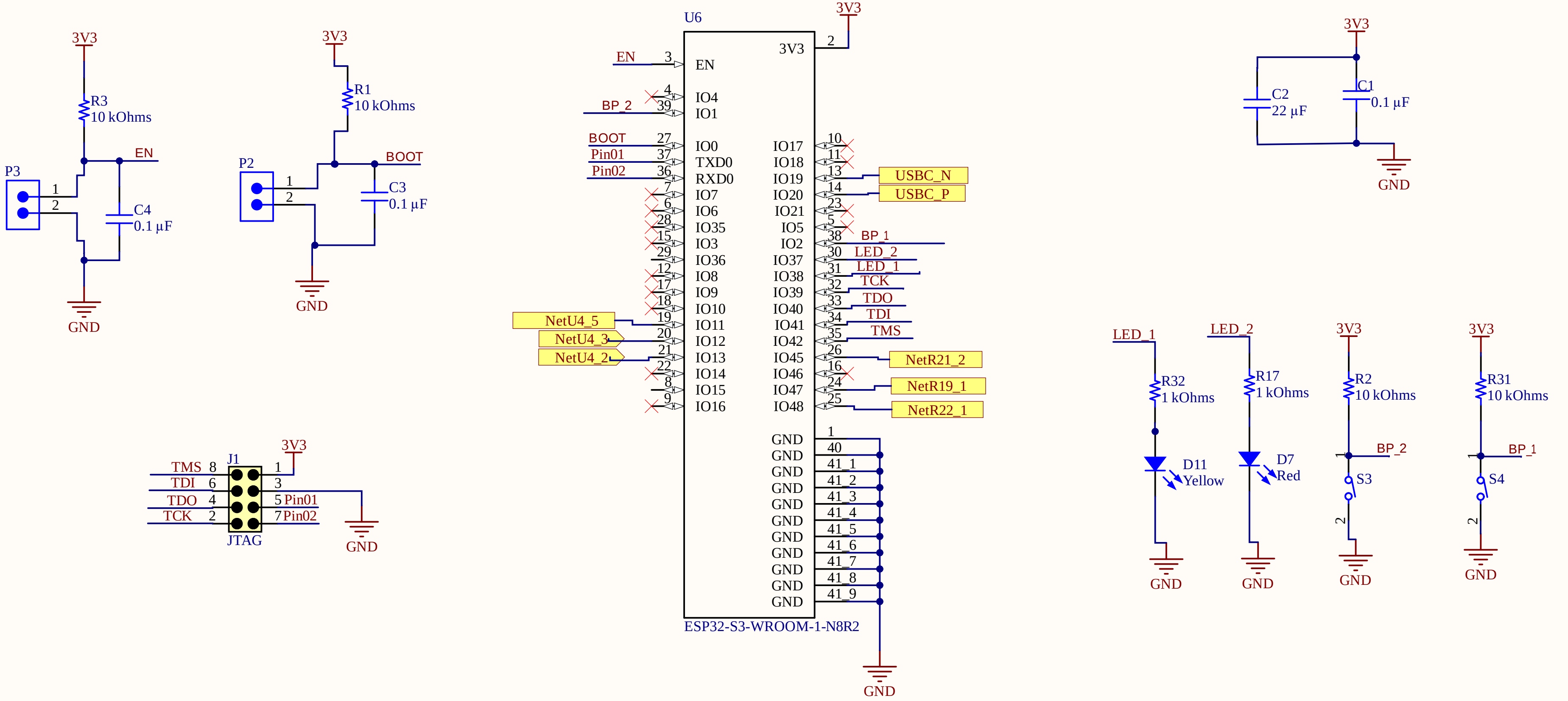

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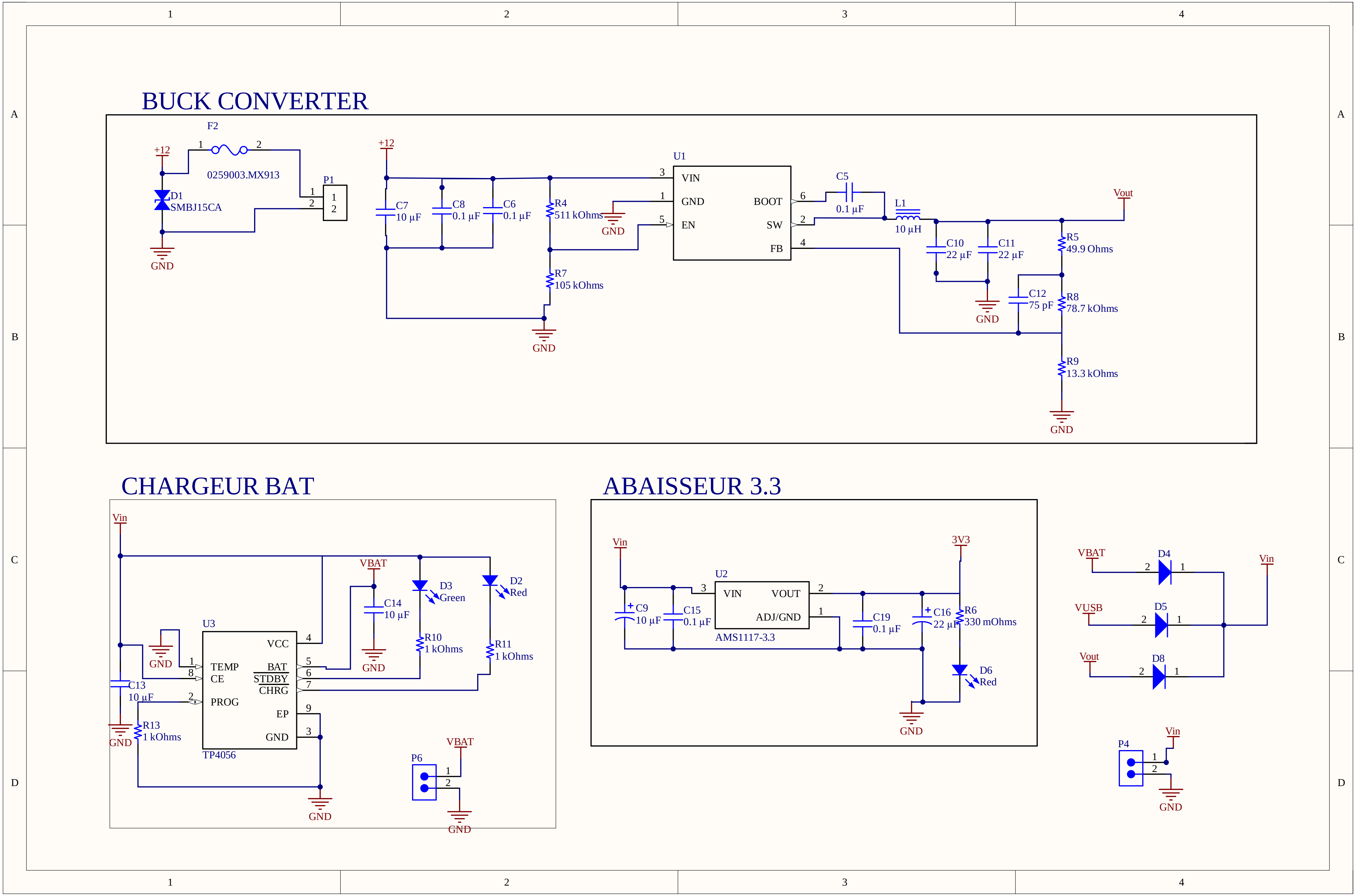
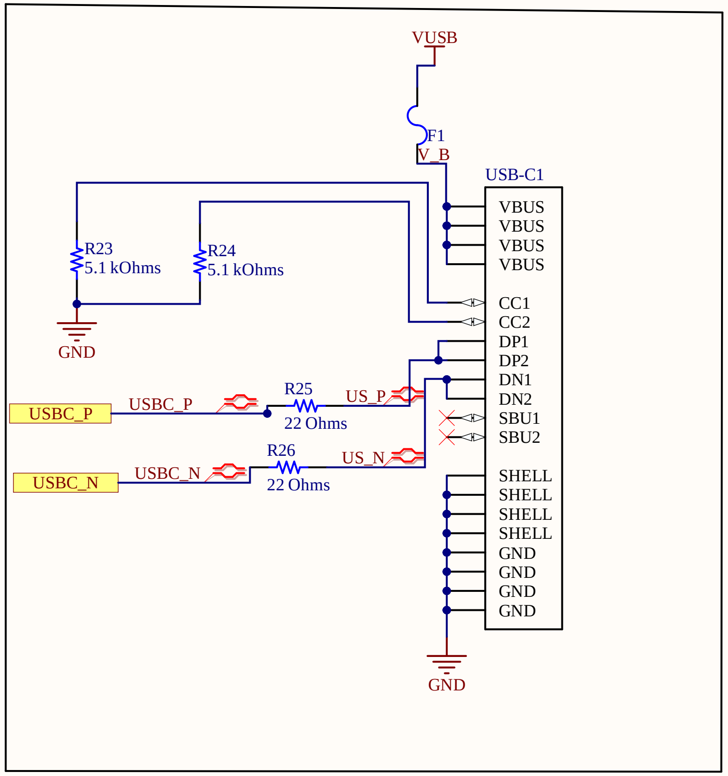
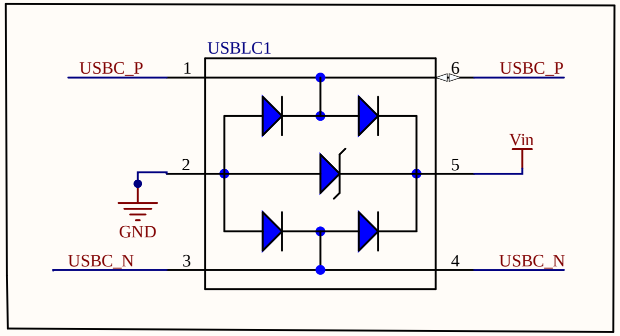

BUCK CONVERTER
F2
1
2
+12
0259003.MX913
D1
SMBJ15CA
GND
P1
1
2
+12
C7
10 µF
C8
0.1 µF
C6
0.1 µF
R4
511 kOhms
R7
105 kOhms
GND
U1
3
VIN
1
GND
5
EN
BOOT
6
SW
2
FB
4
C5
0.1 µF
L1
10 µH
C10
22 µF
C11
22 µF
GND
Vout
R5
49.9 Ohms
C12
75 pF
R8
78.7 kOhms
R9
13.3 kOhms
GND
CHARGEUR BAT
Vin
GND
U3
VCC
4
TEMP
1
CE
8
BAT
5
STDBY
6
CHRG
7
PROG
2
EP
9
GND
3
TP4056
C13
10 µF
GND
R13
1 kOhms
VBAT
C14
10 µF
GND
D3
Green
R10
1 kOhms
D2
Red
R11
1 kOhms
GND
P6
1
2
VBAT
GND
ABAISSEUR 3.3
Vin
C9
10 µF
C15
0.1 µF
U2
3
VIN
VOUT
2
ADJ/GND
1
AMS1117-3.3
C19
0.1 µF
C16
22 µF
3V3
R6
330 mOhms
D6
Red
GND
VBAT
D4
VUSB
D5
Vout
D8
Vin
P4
Vin
GND
1
2
3
4
A
B
C
D

USB-C



ESD



Title

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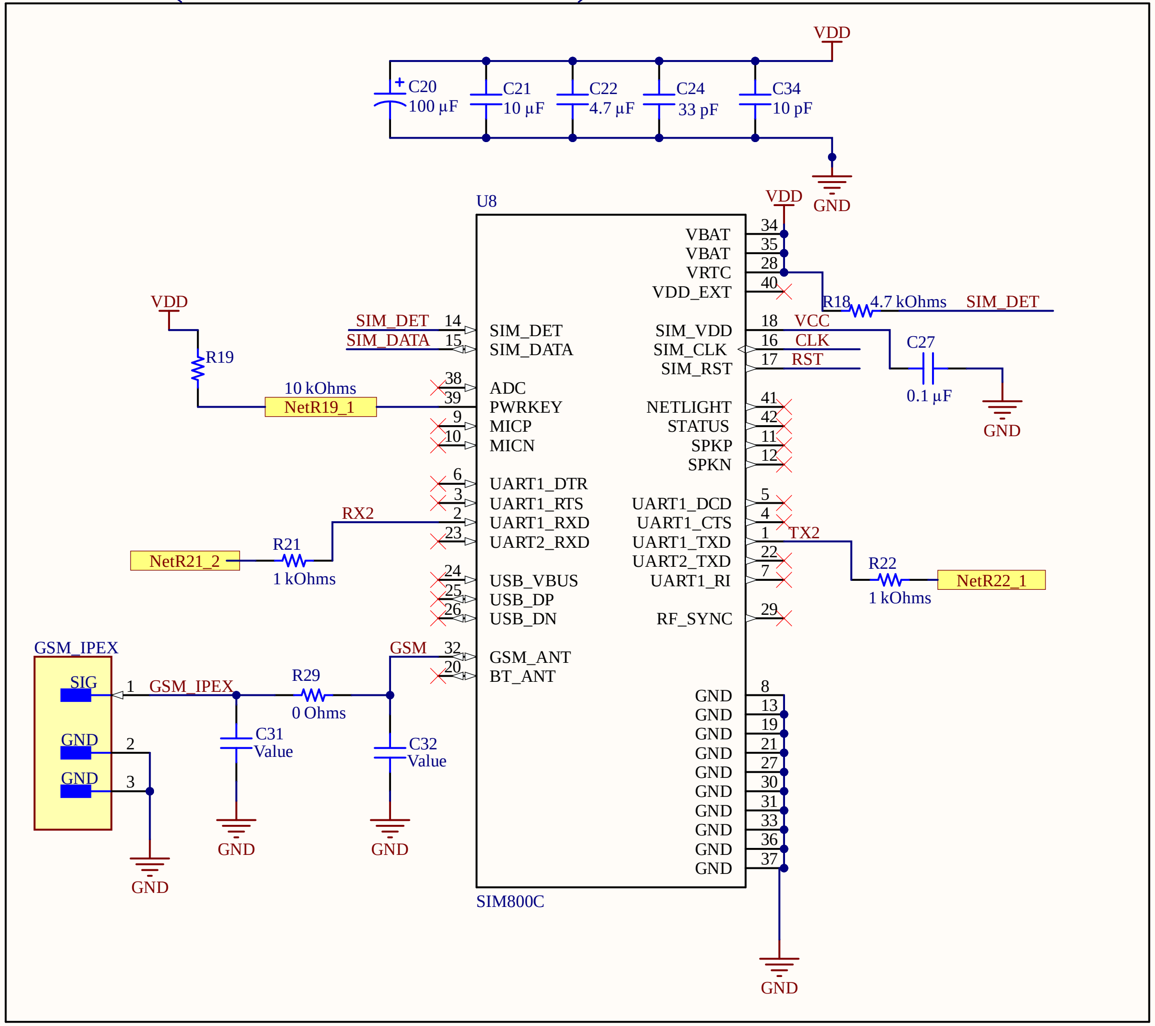
A4

Number

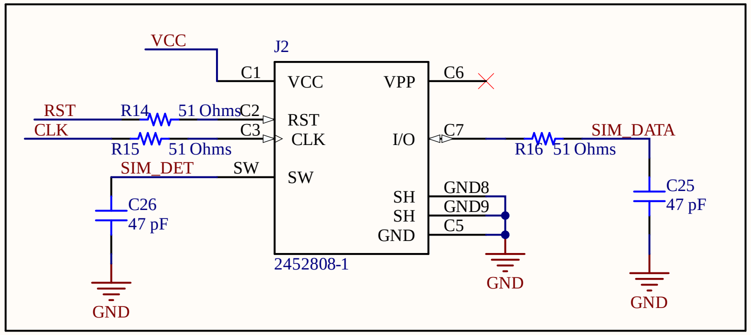
Revision

Date: 10/26/2026
File: D:\Conception PCB\..\USB-C.SchDoc
Sheet of
Drawn By:

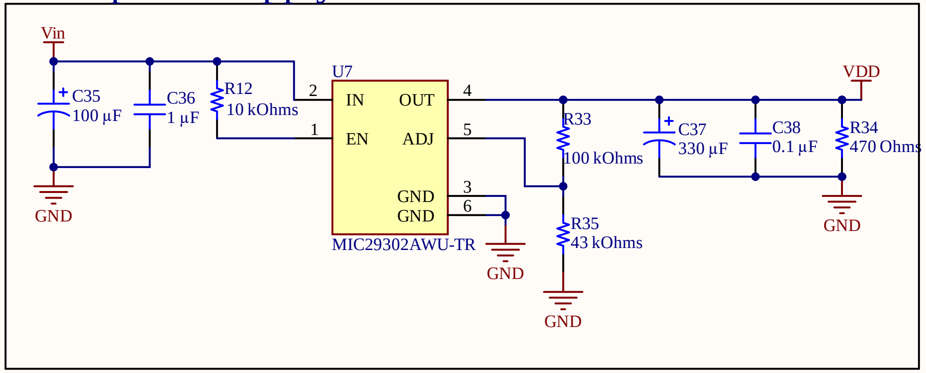
SIM800C(MODULE GSM/GPRS)



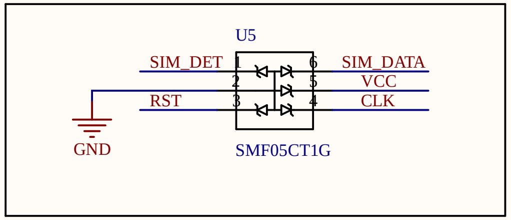
MICRO SIM



LDO power supply



ESD



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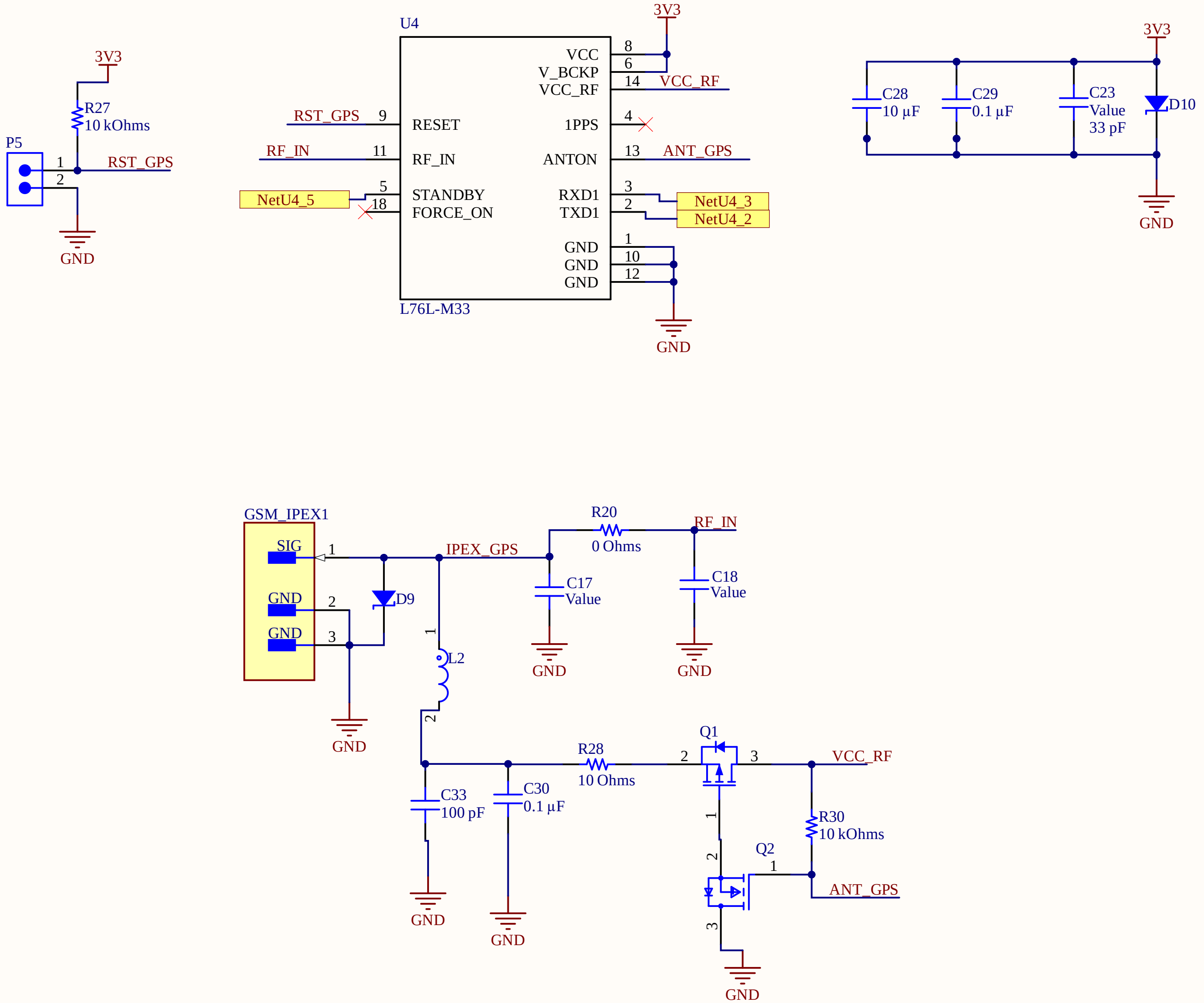

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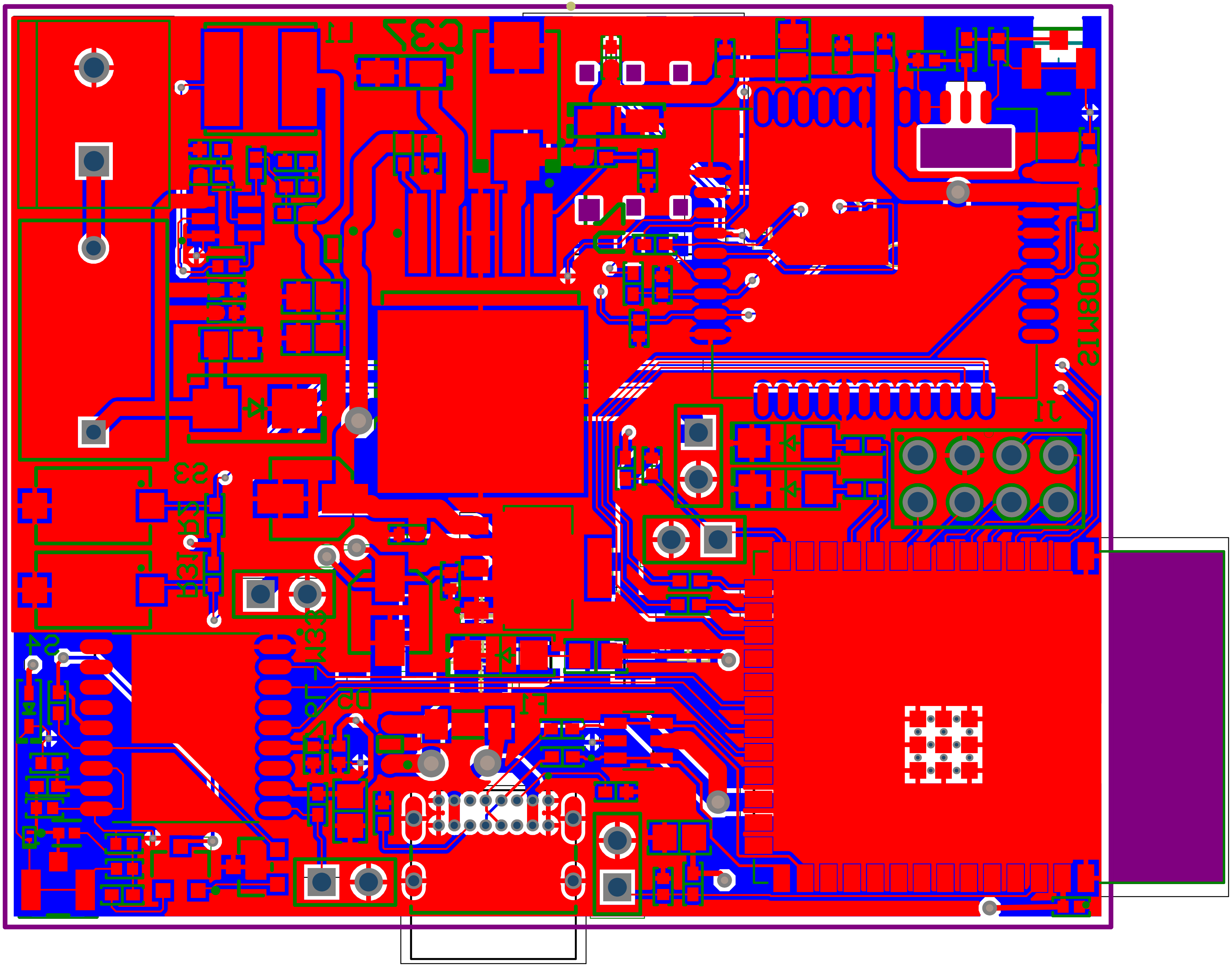
C37
SIM800C
J1
S3
S4
D5
F1

Comment	Description	Designator	Footprint	LibRef	Quantity
0.1 µF	CAP CER 0.1UF 6.3V X5R 0402	C1, C3, C4, C5, C6, C8, C15, C19, C27, C29, C30, C38	CAP 0402_1005	885012105001	12
22 µF	0805 X5R 22UF20% 16V, 0805 X7T 22UF20% 6.3V	C2, C10, C11	CAP 0805_2012	CM21X5R226M16AT, 08056T226MAT2A	3
10 µF	CAP0805 X7S 10UF 10% 6.3V	C7, C13, C14, C21, C28	CAP 0805_2012	GMC21X7S106K6R3NT	5
10 µF	CAP ALUM 10UF 20% 16V SMD	C9	CAP ALSMD 4.3MM H5.4MM	UWX1C100MCL1GB	1
75 pF	CAP0402 COG 75PF 1% 50V	C12	CAP 0402_1005	GMC04CG750F50NT	1
22 µF	CAP ALUM 22UF 20% 6.3V SMD	C16	CAP ALSMD 4.3MM H5.4MM	EEE-0JA220SR	1
Value	CAP SMT04021F6.3VDC, CAP0402 COG 33PF 1% 50V	C17, C18, C23, C24, C31, C32	CAP 0402_1005	NMC0402X5R105K6.3TRPF, GMC04CG330F50NT	6
100 µF	CAP TANT 100UF 20% 4V 1206	C20, C35	CAP POL 1206_3216	T490A107M004ATE500	2
4.7 µF	0402 6.3 V X7T 4.7UF 10%	C22	CAP 0402_1005	KAM05CT70J475KH	1
47 pF	CAP0402 COG 47PF 1% 50V	C25, C26	CAP 0402_1005	GMC04CG470F50NT	2
100 pF	CAP0402 COG 100PF 1% 16V	C33	CAP 0402_1005	GMC04CG101F16NT	1
10 pF	CAP0402 COG 10PF 1% 16V	C34	CAP 0402_1005	GMC04CG100F16NT	1
1 µF	0402 X5R 1UF10% 35V	C36	CAP 0402_1005	0402DD105KAT2A	1
330 µF	330UF (MICRO FARAD)25MOHM @ 100K	C37	CAP POL 2917_7343	TCME337M016R0025E	1
SMBJ15CA	TVS DIODE 15VWM 24.4VC DO214AA	D1	DO214AA SMB	SMBJ15CA	1
Red	LED RED CLEAR 1206 SMD	D2, D6, D7	LED 1206_3216 RED	SML-LX1206SRC-TR	3
Green	LED GREEN CLEAR 1206 SMD	D3	LED 1206_3216 GREEN	IN-S126ATG	1
NSR0320MW2T1G	Schottky Barrier Diode, 2-Pin SOD-323, Pb-Free, Tape and Reel	D4, D5, D8	ONSC-SOD-323-2-477-02_V	CMP-1055-00466-1	3
PESD5V0S1USF,315	TVS DIODE 5VWM 11VC DSN0603-2	D9	NEXPERIA SOD962-2 DSN0603-2	PESD5V0S1USF,315	1
ESD5Z3.3T5G	TVS DIODE 3.3VWM 14.1VC SOD523	D10	SOD523F	ESD5Z3.3T5G	1
Yellow	LED YELLOW CLEAR 1206 SMD	D11	LED 1206_3216 YELLOW	CMD11-21VYC/TR8	1
MFU1206FF00750P100	FUSE BOARD MNT 750MA 63VDC 1206	F1	FUSE 1206_3216 0.6MM	MFU1206FF00750P100	1
0259003.MX913	FUSE BRD MNT 3A 125VAC/VDC RAD	F2	FP-0259-MFG	CMP-08605-000087-1	1
U.FL-R-SMT-1(10)	CONN U.FL RCPT STR 50 OHM SMD	GSM_IPEX, GSM_IPEX1	FP-U_FL-R-SMT-1_10-MFG	CMP-2000-05152-3	2
JTAG	Connector Header Through Hole 8 position 0.100 (2.54mm)	J1	61300821121	6130XX21121_61300821121	1
2452808-1	8 (6 + 2) Position Card Connector NANO SIM Surface Mount, Right Angle Gold	J2	*TE_2452808-1_2452808-1(Primary)	2452808-1	1
10 µH	FIXED IND 10UH 3.4A 61 MOHM SMD	L1	TDK VLS6045EX	VLS6045EX-100M	1
LQG15HS47NJ02J	Multilayer type RF Inductor 47nH ±5% 0.72Ω 300mA 0402	L2	FP-LQG15HH_02-IPC_B	CMP-06042-003943-1	1
282837-2	Terminal Block Connector Wire Receptacle 2 5.08 mm 30-16 AWG Green 13.5 A 250 V | TE Connectivity 282837-2	P1	*TE_282837-2	Bornier 2 vis 5.08mm	1
MTLW-102-05-G-S-172	Low Profile .025" SQ Post Header, Through-hole, Vertical, -55 to 125 degC, 2.54 mm Pitch, 2-Pin, Male, RoHS	P2, P3, P4, P5, P6	SMTC-MTLW-102-05-X-S-172	CMP-1024-00064-1	5
AO3401A	P-Channel MOSFET, 30 V, 4 A, -55 to 150 degC, 3-Pin SOT-23-3L, RoHS, Tape and Reel	Q1	AO-SOT-23-3L_V	CMP-2000-05778-1	1
AO3400A	MOSFET N-CH 30V 5.7A SOT23	Q2	FP-SOT23-MFG	CMP-83955-000006-1	1
10 kOhms	TNPU0402 10K 0.1% T-16 EP1 E3	R1, R2, R3, R12, R19, R27, R30, R31	RES 0402_1005	TNPU040210K0BZEP00	8
511 kOhms	CPF 0402 511K 0.1% 25PPM 1K RL	R4	RES 0402_1005	CPF0402B511KE1	1
49.9 Ohms	RES SMD 49.9OHM 0.05% 1/16W 0402	R5	RES 0402_1005	RG1005N-49R9-W-T1	1
330 mOhms	RES 0.33 OHM 1% 1/3W 0805	R6	RES 0805_2012	RL1220S-R33-F	1
105 kOhms	RES 105 KOHM 0.1% 1/16W 0402	R7	RES 0402_1005	CPF0402B105KE1	1
78.7 kOhms	RCG0402 100 78K7 1% ED	R8	RES 0402_1005	RCG040278K7FKED	1
13.3 kOhms	RES 13.3 KOHMS 0.1% 1/16W 0402	R9	RES 0402_1005	RQ73C1E13K3BTD	1
1 kOhms	TNPU0402 1K0 0.1% T-16 EP1 E3	R10, R11, R13, R17, R21, R22, R32	RES 0402_1005	TNPU04021K00BZEP00	7
51 Ohms	RES SMD 51 OHM 5% 1/16W 0402	R14, R15, R16	RES 0402_1005	CR0402-JW-510GLF	3
4.7 kOhms	RES THIN FILM 0.10% 1/20W 4.7K	R18	RES 0402_1005	NT0402BRD074K7L	1
0 Ohms	RESA-AS 0402 0R JUMPER 63MW	R20, R29	RES 0402_1005	CR0402AJ/-000GAS	2
5.1 kOhms	TNPU0402 5K1 0.1% T-16 EP1 E3	R23, R24	RES 0402_1005	TNPU04025K10BZEP00	2
22 Ohms	RES 22 OHM 1% 1/8W 0402	R25, R26	RES 0402_1005	RC0402FR-7W22RL	2
10 Ohms	TNPU0402 10K 0.1% T-16 EP1 E3	R28	RES 0402_1005	TNPU040210K0BZEP00	1
100 kOhms	TNPU0402 100K 0.1% T-16 EP1 E3	R33	RES 0402_1005	TNPU0402100KBZEP00	1
470 Ohms	RES SMD 470 OHM 5% 1/16W 0402	R34	RES 0402_1005	CR0402-JW-471GLF	1
43 kOhms	TNPW0402 43K 0.1% T-13 EP1 E3	R35	RES 0402_1005	TNPW040243K0BYEP	1
434123025816	Tactile Switch SPST-NO Top Actuated Surface Mount	S3, S4	434123025816	434123025816	2
	4.5-V to 28-V Input, 3-A Output, EMI Friendly Synchronous Step-Down Converter	U1	TPS54302DDCR	TPS54302DDCR_1	1
AMS1117-3.3	Linear Voltage Regulator IC Positive Fixed 1 Output 1A SOT-223-3L	U2	AMS1117 3.3V	AMS1117-3.3	1
TP4056	Complete single cell Li-Ion battery with a constant current / constant voltage linear charger	U3	TP4056	TP4056	1
L76L-M33	L76-L is a concurrent receiver module integrating GPS, GLONASS, Galileo and QZSS systems. With 33 tracking channels, 99 acquisition channels and 210 PRN channels, L76-L can acquire and track any mix of GPS, GLONASS and SBAS signals. Designed to be compatible with Quectel L76 module in the compact and unified form factor, L76-L provides a built-in LNA for better performance in weak signal areas.	U4	*XCVR_L76L-M33	L76L-M33	1
SMF05CT1G	12.5V Clamp 8A (8/20Âµs) Ipp Tvs Diode Surface Mount SC-88/SC70-6/SOT-363	U5	*SOT65P210X110-6N	SMF05CT1G	1
ESP32-S3-WROOM-1-N8R2	WiFi 802.11a/b/g/n, Bluetooth v5.0 Transceiver Module 2.4GHz Antenna Not Included -	U6	*XCVR_ESP32-S3-WROOM-1-N8R2_ESP32-S3-WROOM-1-N8R2(Primary)	ESP32-S3-WROOM-1-N8R2	1
MIC29302AWU-TR	IC REG LINEAR POS ADJ 3A TO263-5	U7	FP-T0263-5LD-PL-1-MFG	CMP-06324-000015-1	1
SIM800C	Module: GSM/Bluetooth; GPRS; 2G; 1800MHz, 900MHz; 85.6kbps; SMD	U8	SIM800C	SIM800C	1
USB4085-GF-A_REVA4	USB-C (USB TYPE-C) USB 2.0 Receptacle Connector 24 (16+8 Dummy) Position Through Hole, Right Angle, low cost, 3.46mm profile, 16 pin, horizontal, Top mount	USB-C1	Conn Type C 2.0	USB4085-GF-A_REVA4	1
USBLC6-2SC6	17V Clamp 5A (8/20Âµs) Ipp Tvs Diode Surface Mount SOT-23-6	USBLC1	USBLC6-2SC6	USBLC6-2SC6	1